

8. (a) Aluminium reacts with oxygen to form aluminium oxide.

Using outer electrons only, draw a dot and cross diagram to show the bonding in aluminium oxide.

[2]

- (b) Aluminium also reacts with chlorine to form aluminium chloride.

A 0.400 g sample of aluminium chloride was heated to 220 °C. The vapour produced occupied a volume of 60.8 cm³ at a pressure of 101 kPa.

Show that the molecular formula of aluminium chloride in the vapour is Al₂Cl₆.

[4]

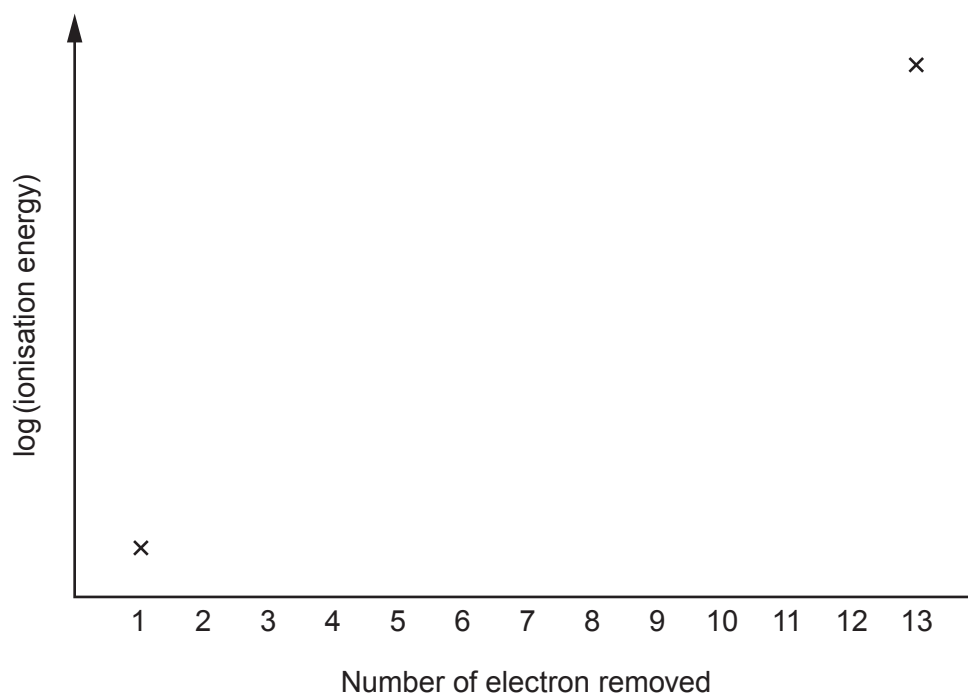
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- (c) (i) Sketch a graph of $\log(\text{ionisation energy})$ for the successive ionisations of aluminium.

The first and last points have been plotted for you.

[1]



- (ii) I. Explain the general slope of the graph.

[1]

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- II. Explain the reason for any sharp changes in the graph.

[1]

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- (d) Although aluminium has 25 known isotopes only two of them occur naturally. These are ^{27}Al which is stable and ^{26}Al which is radioactive.

^{26}Al decays by electron capture and its half-life is 7.2×10^5 years.

- (i) Give the mass number and symbol of the species produced when an atom of ^{26}Al decays. [1]

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- (ii) If 8.0 mg of ^{26}Al decays by electron capture, calculate the mass in **grams** of ^{26}Al left after 2.88×10^6 years. [2]

Mass = g

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